DATA ITEM DESCRIPTION

Title: MANUFACTURING READINESS LEVEL (MRL) ASSESSMENT REPORTING

Number: DI-SESS-81974A
Approval Date: 20260810
AMSC Number: 10687
Limitation: N/A
DTIC Applicable: No
GIDEP Applicable: No
Preparing Activity: 11 (AFRL/REM)
Project Number: SESS-2026-006
Applicable Forms: Example MMP Template_20260610

(Copies of Example MMP Template_20260610) are available at dodmrl.com or dodmrl.org.

Overview / Use:
This Data Item Description (DID) is applicable to industry, contractors, and government organizations and supports use across all acquisition pathways and lifecycle phases, from early design and development through production. It defines the minimum required content necessary for documenting results of a Manufacturing Readiness Level (MRL) Assessment and to support planning for subsequent manufacturing maturation activities. The content defined herein may be adjusted to align with program scope, complexity, and maturity, and use of this DID is typically specified within program tasking and the Contract Data Requirements List (CDRL).

Following completion of an MRL Assessment—also referred to as a Manufacturing Readiness Assessment (MRA)—two primary documents are produced: the *MRL Assessment Report and the Manufacturing Maturation Plan (MMP).*

- The MRL Assessment Report documents the current state of manufacturing maturity, including assessed MRLs, supporting artifacts, identified gaps and shortfalls, and associated manufacturing risks.
- The MMP builds upon these results by defining structured manufacturing maturation activities to address gaps, mitigate risks, and advance manufacturing maturity.

These documents are intended to be used together to provide a clear and traceable linkage between assessment results and execution of manufacturing maturation efforts. They support risk-informed decision-making and enable progression of manufacturing maturity across applicable MRL threads, subthreads, and associated criteria. This DID supersedes DI-SESS-81974.

Requirements:
1. Following the completion of an MRL Assessment, the organization that leads the assessment shall generate an MRL Assessment Report and provide the report to the organization that they have assessed. The assessed organization shall use this report to develop the MMP, when required.
2. The organization that leads the MRL Assessment should define the following for an MRL Assessment Report and an MMP:
   a. additional reporting requirements,
   b. content tailoring for program applicability, and
   c. the required deliverable format.

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3. Execution of MMP activities should be coordinated between both organizations that took part in the assessment.
4. *MRL Assessment Report* – The report shall include the following content:
   a. Description of the technologies, components, items, subsystems, and systems assessed, including the current state of manufacturing capabilities.
   b. Program objectives and requirements, including production rate targets, schedule constraints, and MRL targets.
   c. Personnel involved in the MRL Assessment from both the assessed organization and the responsible organization.
   d. Dates, locations, and descriptions visits conducted as part of the assessment.
   e. Description of how the MRL Assessment was conducted, including the MRL Criteria Matrix and Deskbook version used, the extent to which criteria were assessed, and any tools or methodologies applied; if any criteria were tailored or not assessed, document the approach and rationale.
   f. Key and critical manufacturing technologies and processes associated with each assessed technology, component, item, subsystem, and system.
   g. Assessed MRL value for each applicable thread and subthread within the MRL Criteria Matrix, including description of supporting artifacts used to satisfy criteria.
   h. Manufacturing maturity gaps, shortfalls, and contributing factors related to MRL thread and subthread criteria, including required actions and artifacts to achieve target MRLs.
   i. Summary of identified risks to the program and the respective risk likelihood and consequences. Ensure risk is assessed against cost, schedule, and performance objectives.

5. *Manufacturing Maturation Plan (MMP)* – The MMP defines distinct activities that address identified manufacturing maturity gaps, shortfalls, and risk areas. These activities are intended to promote manufacturing maturity and readiness across applicable MRL threads, subthreads, and associated criteria. MMP activities should be structured at a level appropriate for execution, tracking, and management. An example MMP template can be found at dodmrl.com or dodmrl.org. Each MMP shall include the following content:
   a. The applicable technologies, items, components, subsystems, or systems, and indicate whether each activity supports a new task, is on the critical path, or represents an alternate path.
   b. The manufacturing maturity gaps, shortfalls, and risk factors addressed by the activity, its benefits to the product/program, and any consequences if not executed
   c. The course of actions, including specific tasks required to execute the activity
   d. Deliverables and expected outcomes, including the targeted MRL upon completion (overall MRL or specific threads and subthreads).
   e. Results, artifacts, and criteria or metrics used to evaluate success and demonstrate advancement in manufacturing maturity.

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f. Whether the activity is within or outside the scope of an existing program; for activities outside the scope, a Rough Order of Magnitude (ROM) cost estimate to indicate level of effort.
g. Assigned personnel and responsible organizations for execution, as well as authorities responsible for review and approval.
h. ROM schedule and timeline, including key milestones and deliverables, and an assessment of alignment with product delivery requirements.
i. Potential challenges, constraints, and risks associated with execution, including impacts to required resources (for example, personnel, equipment, and facilities).

References: The latest versions can be found at dodmrl.com or dodmrl.org.

- MRL_Criteria Matrix_V2025_R1
- MRL Deskbook V2025 July
- DoD Risk, Issue, and Opportunity (RIO) Management Guide (2023)

End of DI-SESS-81974A

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